KawruhBhumi Lab

PRACTICAL USER GUIDE · EDITION 1.1 · OCTOBER 2026

Use the lab with confidence.

A complete, method-by-method guide to KawruhBhumi Lab — from your first scenario to a defensible interpretation.

7 methods | 35 theory chapters | 21 guided experiments | 22 pages | EN / ID

01
Understand
Read the physics and the assumptions.

02
Simulate
Change one control and run a validated solver.

03
Interpret
Explain the evidence, the alternative, and the limit.

METHOD CHAPTERS
One page per solver
Controls, charts, best practice, limits, exercise.

SIMULATION MATRIX
Choose with evidence
Seven solvers compared by question and limit.

CLASSROOM & EXPORT
Teach and assess
55-minute plan, rubric, evidence pack.

MOBILE & SUPPORT
Use it anywhere
390px workflow, troubleshooting, glossary.

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Teaching tool only — not a calibrated instrument.
Every synthetic result states its approximations.

CONTENTS

What is inside

Read it in order, or open the page for the method you need. The method chapters are the heart of this guide.

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How to read this guide

- Pages 3–8 build the habits. Pages 9–16 explain each simulation. Pages 17–22 cover classroom, export, mobile, troubleshooting and vocabulary.
- Each method page follows the same pattern: what you control, what you read, best practice, limits, and one exercise.
- Screenshots follow the selected language. The English and Indonesian editions have identical structure.

Conventions

- Routes are written as `#/simulator/gpr`. Open them from the navigation bar or type them after the site address.
- **Synthetic** means computed from a model — never measured field data.
- Solver badges (ANALYTICAL, APPROXIMATE, NUMERICAL) tell you how the response is computed. Read the solver note before interpreting.

STUDENT

Learn the loop

Pages 3–6, then your method page, then compare and challenge (17). Finish with the evidence pack (19).

TEACHER

Plan a lesson

Read the method pages and classroom plan (18), download the teacher pack, and choose one experiment per group.

REVIEWER

Check the physics

Read the assumptions, solver notes and limits on pages 9–16. Use the glossary (22) for symbols.

A NOTE ON HONESTY

Every page asks you to state what the model does *not* include. That habit is the difference between a demo and an interpretation.

New in edition 1.1

✓ A dedicated page for each of the seven simulations, not only HVSR.

✓ Best-practice workflow, preset guidance and a lab-notebook habit.

✓ Logo and wordmark on the cover and in every page header.

✓ A seven-method simulation matrix and a separate HVSR inversion page.

✓ Classroom plan, evidence pack, mobile workflow and troubleshooting.

✓ Localized screenshots: the EN edition shows EN, the ID edition shows ID.

START HERE

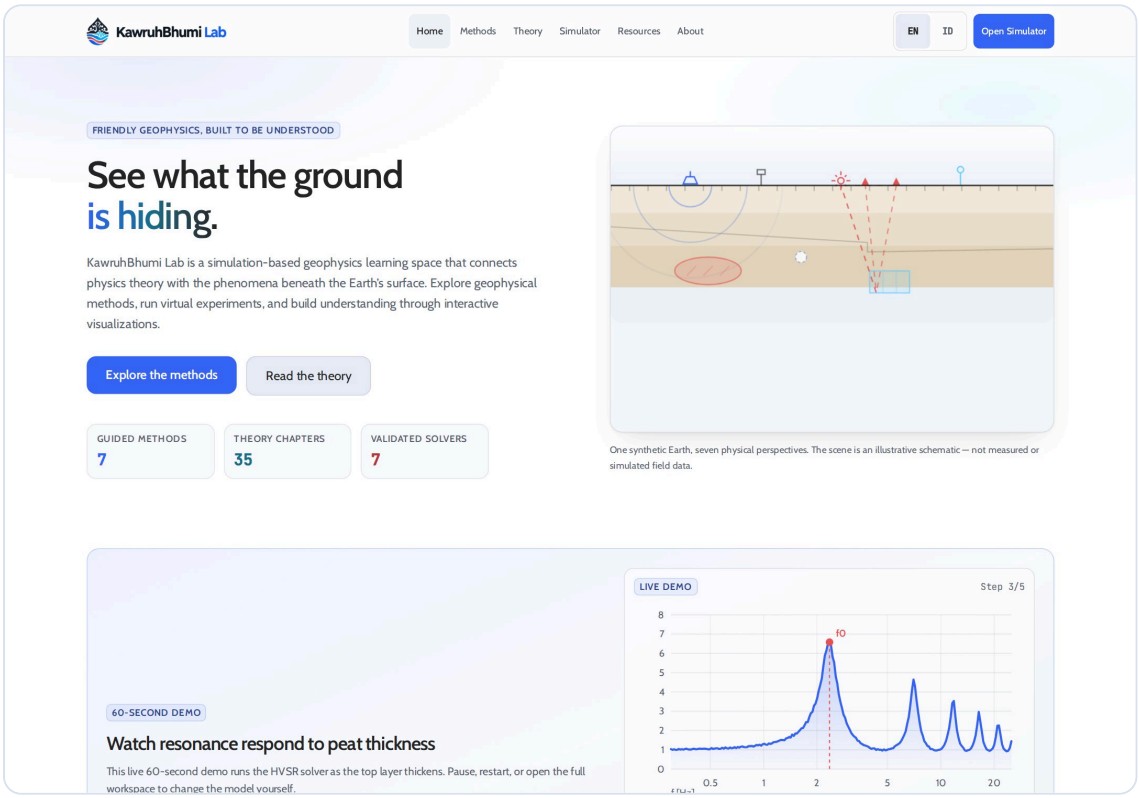
Your first 20 minutes

You do not need to read all 35 chapters before your first run. Start with one method, one preset, and one question.

1. **Choose a method**
   Open `#/metode`. Read "What is measured" and "Why it works" before opening the workspace.
2. **Read one chapter**
   Open `#/teori`. Start with chapter 1 if the symbols are unfamiliar.
3. **Load an Easy preset**
   In `#/simulator/<method>`, choose an Easy card. Read its expected observation.
4. **Read the solver note**
   Find the classification and assumptions panel. Note the dimensions, units, and what is not modelled.
5. **Write a prediction**
   One sentence: "If I increase X, Y should…" State the physical reason.
6. **Run the solver**
   Press **Run Simulation**. Wait for the chart to settle before reading it.
7. **Change one control**
   Keep everything else fixed. Run again. Use Compare or Sensitivity to isolate the effect.
8. **Record the evidence**
   Save the permalink, export CSV/PNG, and write one limitation sentence.

BEST PRACTICE

Predict before you run. A prediction turns a demo into a test of your mental model. Being wrong is useful evidence — write down why.



Home page: seven methods, 35 chapters, and a live HVSR demo. Use the EN / ID toggle at any time.

The five-minute version

1. Open `#/simulator/gpr`.
2. Load "Dry sand over bedrock" (EASY).
3. Press **Run Simulation**.
4. Read the radargram and one diagnostic.
5. Change frequency once, run again, and explain the difference.

You are ready when…

- You can name the measured property.
- You can find the solver note.
- You can explain one axis and its unit.
- You changed one control and predicted the effect.
- You saved a permalink or export.

ORIENTATION

A small library, not one long page

Each door has a different job. Knowing which door to use saves time and prevents “chart-first” interpretation.

DOOR	ROUTE	WHAT IT IS FOR
Home	`#/`	Orientation, live HVSR demo, guide downloads, and entry points.
Methods	`#/metode`	Seven profiles: measured property, mechanism, applications.
Theory	`#/teori`	35 chapters — five per method: measurement, physics, model, example, limits.
Simulator	`#/simulator`	Seven workspaces with presets, controls, charts, compare/sweep, challenge, report.
Resources	`#/resources`	21 guided experiments, glossary, formula sheet, EN/ID teacher pack.
About / Disclaimer	`#/about · #/disclaimer`	Scope, assumptions, educational limits, contact.

STUDENT PATH

Concept → practice

Methods → Theory chapter 1 → Simulator Easy → Compare A/B → Challenge → Report. One method per session.

TEACHER PATH

Prepare → assess

Resources teacher pack → choose experiment → project workspace → worksheet → assess permalink, screenshot, CSV/PDF, limitation.

REVIEWER PATH

Assumption → test

Theory limits → Simulator Advanced → Sensitivity sweep → Export → write the alternative model and uncertainty.

WHICH DOOR FIRST?

If the symbols are unfamiliar, start with Theory. If you already know the physics, start with Simulator. In both cases, read the assumptions before interpreting a chart.

Choose a path by time budget

TIME	PATH	WHAT YOU SHOULD BE ABLE TO DO
15 min	Home → one method → Easy preset → Run	Read the chart and name one assumption.
45 min	Theory chapter 1 → Simulator → Compare A/B	Predict one change and explain the delta.
2 hours	Method page → Theory 1–3 → Sweep → Challenge → Report	Defend an interpretation and its limits.
One term	One method per week; 21 experiments across seven methods	Build a portfolio of evidence packs.

Home is not the lesson

The live HVSR demo is a hook. The lesson is in the workspace, where you control the model and read the solver note.

Theory is not optional

A chart without the measured property and the model assumptions is decoration. Read at least chapter 1 and chapter 5.

Resources is the bridge

Use the 21 experiments and the teacher pack to turn a demo into an assignment with evidence.

WORKFLOW

The same six-step loop in every workspace

Predict → baseline → one variable → run → explain → record. It keeps comparisons honest and interpretations defensible.

1 Predict

Write the expected direction and approximate magnitude before pressing Run. State the physical reason.

2 Set a baseline

Load a preset. Record the scenario name and the values you will change. Pin the baseline if Compare is available.

3 Change one variable

Keep all other controls fixed. Use Compare A/B or a one-factor sensitivity sweep to isolate the effect.

4 Run and read

Check warnings, axes, units, and assumptions before reading the shape of the curve.

5 Explain

Describe what changed, why the physics predicts it, and one alternative model that could fit the same observation.

6 Record

Save the permalink, export CSV/PNG, print the report, and write a two-sentence limitation statement.

A USEFUL HABIT

If you cannot explain the direction of the change before running, you are exploring — and that is fine. Label it as exploration, then turn it into a test.

Three traps to avoid

Changing several controls

You cannot attribute the result to one cause. Change one thing, then run.

Comparing different scenarios

Preset A and preset B may differ in many parameters. Use the same scenario for a controlled test.

Reporting only the best fit

A single curve hides non-uniqueness. Name one alternative model that could also fit.

Worked example: GPR frequency

1. Load “Buried pipe in moist soil”. Run and pin as A.
2. Predict: 900 MHz should sharpen the wavelet but attenuate faster than 400 MHz.
3. Change only antenna frequency. Run and pin as B.
4. Read the resolution and amplitude deltas. Check the attenuation warning.
5. Write: “Sharper in dry sand, but the wet-clay model loses the deeper event.”

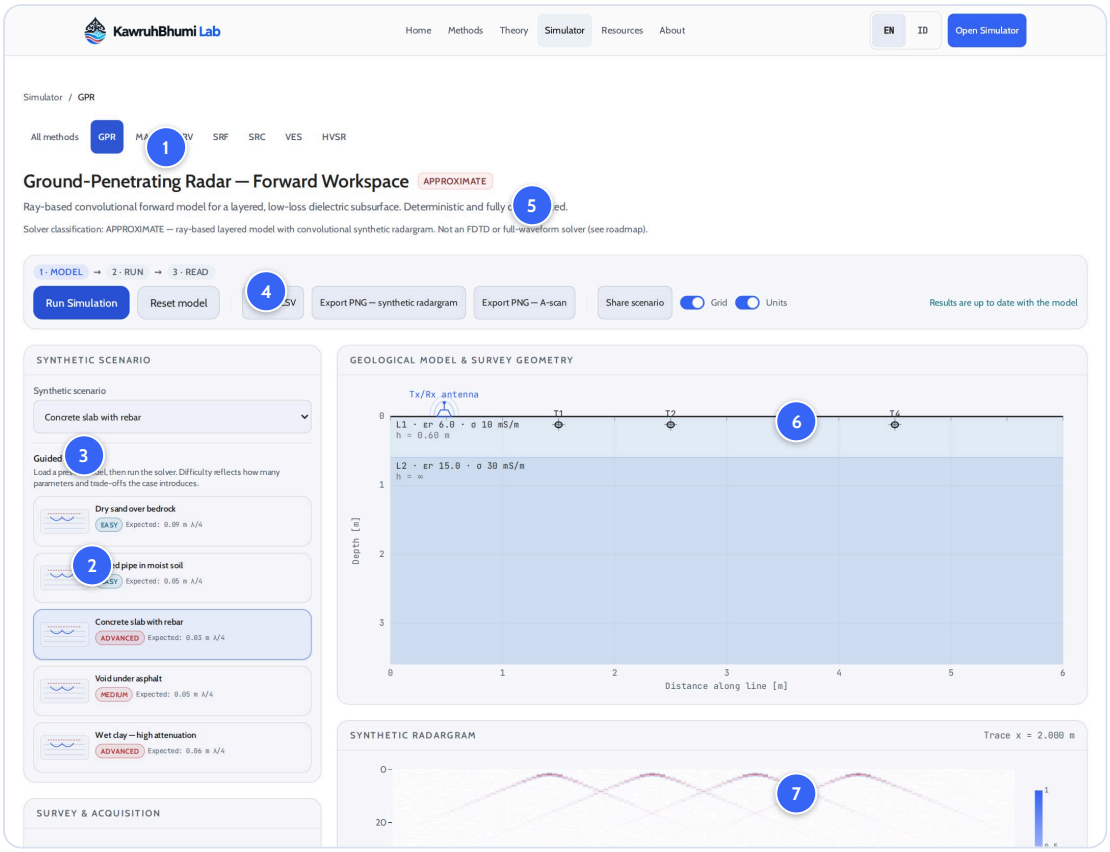
Lab notebook template

FIELD	EXAMPLE
Question	Does frequency trade resolution for penetration?
Baseline	Buried pipe, 400 MHz, run A
Change	Frequency 400 → 900 MHz only
Prediction	Sharper, shallower
Observation	Pulse narrower; deep event weaker
Limit	2D ray model; no antenna coupling

READING THE SCREEN

Anatomy of a workspace

The same layout appears across the seven methods. Learn it once, then spend your attention on the physics.



The GPR workspace. The same anatomy repeats in all seven methods; panels below the fold are reached by scrolling.

1 **Method switcher & breadcrumb.** Move between methods without losing your place.

2 **Preset cards.** Difficulty and expected observation. Start Easy.

3 **Model & survey controls.** Geometry, properties, acquisition, advanced settings.

4 **Run / Reset / step bar.** Model → Run → Read. Reset restores the selected scenario.

5 **Solver note.** Classification, dimensionality, and what is not modelled.

6 **Model cross-section.** The geometry the solver actually sees.

7 **Synthetic response.** Radargram, profile, curve, gather, or seismogram. Use the chart toolbar for zoom, CSV and PNG.

8 **Below the fold.** Diagnostics, warnings, compare/sweep, challenge, report, and assumptions.

READ THIS FIRST

A warning is not decoration. Fix the input or acknowledge the limitation before interpreting the chart.

Panel finder

WHAT YOU NEED	WHERE IT IS	WHY IT MATTERS
Change the geological model	Model & survey controls (left)	The solver only sees what you enter here.
Read the synthetic response	Main chart panel (right)	This is the model response, not field data.
Check units and warnings	Below the chart	Units and warnings bound the interpretation.
Compare two runs	Compare / Sweep panels	Isolates one control and shows the delta.
Test your understanding	Challenge panel	Scores curve similarity, not geological truth.
Export evidence	Chart toolbar + Report	CSV/PNG/PDF carry the model and assumptions.

SCENARIOS

Start from a known model

Presets are teaching cases with a documented model, difficulty and expected observation. Use them to learn the controls before editing freely.

EASY

Learn the layout

Few parameters and a stable response. Use it to learn the controls and the chart.

MEDIUM

Test a trade-off

Two or three interacting parameters. Predict before you change.

ADVANCED

Confront ambiguity

More parameters and stronger non-uniqueness. Justify your choice.

How to use a preset

- **Load it.** The model, survey settings and expected observation change together.
- **Read the expected card.** It tells you what the synthetic response should show.
- **Predict, then run.** Write one sentence before pressing Run.
- **Change one thing.** Keep the scenario fixed; vary one control.

Scenario codes & permalinks

- The teacher pack uses runnable `?scenario=` codes. Open the full URL to restore the model.
- **Share scenario** copies the full state: method, scenario, model and display settings.
- Include the method and scenario name when you paste a link into a report or LMS.

A PRESET IS A STARTING POINT

It is a known model, not the answer. The learning happens when you change one control, predict the effect, and explain the difference.

What to record for every run

- ✓ Method and scenario name
- ✓ Prediction before running
- ✓ One alternative model
- ✓ Parameters changed (one at a time)
- ✓ Screenshot or CSV/PNG export
- ✓ One limitation sentence

From preset to your own model

1. **Find the nearest case**
   Choose the preset with the same geometry, not the same numbers. A pipe is closer to a cylinder than to a layer.
2. **Copy the structure**
   Keep the layer count or body count. Change only the property you know from your site.
3. **Label the uncertainty**
   Write which values are measured, which are estimated, and which are guesses.
4. **Save and share**
   Use Share scenario, then paste the link into your notebook with the method and scenario name.

Pitfall: preset hopping

Comparing two presets changes many parameters at once. Use the same scenario and change one control.

Pitfall: hidden defaults

Noise, seed and display toggles persist. Check them before blaming the solver.

Pitfall: false precision

A preset is a teaching model. Its numbers are plausible, not site-calibrated.

SIMULATION MATRIX

Seven simulations at a glance

Use this matrix to choose a method, then read its page for the controls, the output, the best practice and the limits.

METHOD	QUESTION IT ANSWERS	WHAT YOU READ	START WITH	MAIN LIMIT
GPR	What lies in the top few metres?	Radargram, A-scan, hyperbola apex	Dry sand; buried pipe	Conductive clay limits penetration
MAG	Where is magnetized or ferrous material?	ΔT profile, anomaly map, peak-to-peak	Buried steel tank; dike	Remanence and regional field omitted
GRV	Where is a density contrast?	g profile, anomaly map, excess mass	Shallow cavity; salt dome	Non-uniqueness; no terrain correction
SRF	What is the layered geometry?	Zero-offset trace, NMO gather, R	Sand–shale; gas sand	Primaries only; no multiples or AVO
SRC	How fast is each layer, and how deep?	t(x) first arrivals, crossover, intercept	Soil over bedrock; weathered layer	Hidden layers and velocity inversions
VES	How does resistivity change with depth?	ρa sounding curve, layer model, equivalence	Two-layer bedrock; clay layer	1D only; equivalence is real
HVSR	What is the site resonance?	H/V curve, f0, A0, Vs ensemble	IKN alluvium; peat column	1D SH proxy; higher modes

Target or void

GPR for the shallowest detail; gravity or magnetics for a broader mass or metal signature.

Layers and depth

Refraction for first-arrival velocities; reflection for continuous interfaces; VES for resistivity layering.

Site resonance

HVSR for f0 and A0, then the labelled inversion for a non-unique Vs ensemble.

THE QUESTION COMES FIRST

A method is not “better” because it looks modern. It is better when its measured property answers your question at the scale you care about. If two methods disagree, that disagreement is evidence — report both and explain it.

Solver badges

ANALYTICAL — closed-form or exact layered solution (gravity, magnetic prisms, reflection, refraction, parts of GPR).

APPROXIMATE — ray or low-loss approximation with stated limits (GPR, magnetic spheres).

NUMERICAL — quadrature or search (VES Hankel integral, HVSR inversion).

One Earth, seven observables

The same synthetic ground can be described by εr and σ, χ, Δρ, Z, Vp, ρ, or resonance. No single number is the ground; each is a projection of it.

When you compare methods, compare the property and the scale — not just the picture.

Two-question shortcut

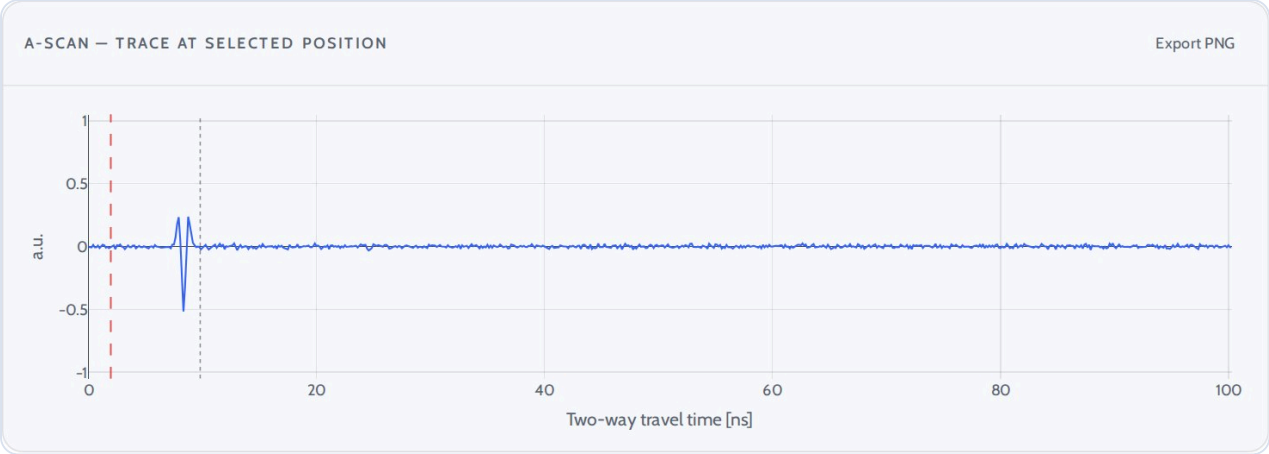
1. Is the target shallow, sharp and high-contrast? Start with GPR or magnetics.

2. Is the question about layering, mass or resonance? Choose seismic, gravity/VES, or HVSR.

METHOD 01 · GPR

Ground-penetrating radar

GPR sends high-frequency electromagnetic pulses into the ground and records reflections where relative permittivity εr or conductivity σ changes. The workspace is a fast ray-based 2D approximation — ideal for learning the trade-offs, not a full-waveform simulator.



A-scan from the concrete-slab preset at x = 2 m (900 MHz). The wavelet is clear; the shared radargram colour scale is fixed at ±1, so weak events are easier to read in the A-scan and diagnostics.

SOLVER	SYNTHETIC OUTPUT	KEY UNITS
APPROXIMATE	Radargram + A-scan	ns · m · MHz

WHAT YOU CONTROL

- **Antenna frequency** (10–3000 MHz): higher frequency improves resolution but shortens penetration.
- **Trace spacing and line length**: lateral sampling versus the 3000-trace guard.
- **Layer stack**: thickness, εr and σ for each finite layer and the half-space.
- **Targets**: point or cylinder, position, depth, radius, εr and σ. The metal option uses a PEC approximation.
- **Noise and seed**; time window; grid, units and PNG/CSV export.

WHAT YOU READ

- **Radargram**: distance along the line versus two-way time. Flat events follow layers; point targets produce hyperbolas.
- **A-scan**: the wavelet at the selected trace. Read arrival time, polarity and whether the event is above the noise.
- **Hyperbola apex** $t_0 \rightarrow$ depth $z = v \cdot t_0/2$ with $v = c/\sqrt{\varepsilon_r}$. The curvature also carries velocity information.
- **Diagnostics**: λ/4 vertical resolution, 1/e amplitude depth, trace count, time window and warnings.
- **Picks table**: model-derived TWT, converted depth and reflection coefficient — aids, not inversion.

BEST PRACTICE

Start in dry sand at 400 MHz and learn the A-scan. Then change only frequency and compare the same target. Keep the line, target and time window fixed so the difference is attributable. If the radargram is pale, use the A-scan and diagnostics rather than increasing noise.

LIMITS TO STATE

2D independent 1D columns; no 3D focusing. Low-loss ray approximation, no multiples, antenna radiation pattern or coupling. Frequency-dependent conductivity, dispersion and full-waveform effects are absent. Metal targets are PEC approximations. Depth is a model conversion, not an inversion.

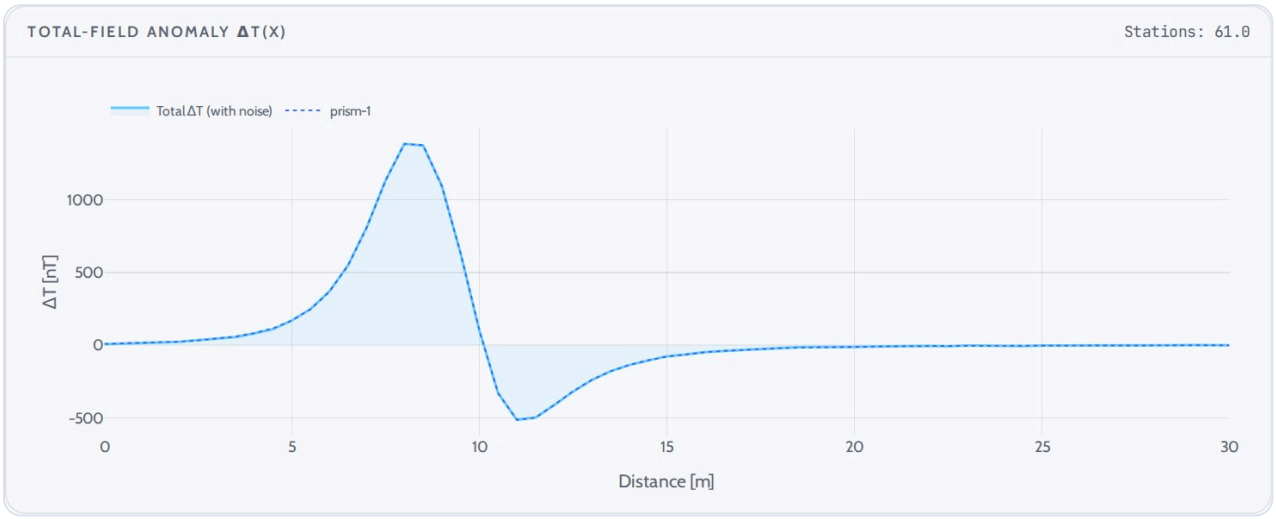
Exercise · Frequency versus depth

Load `#/simulator/gpr?scenario=sandBedrock`. Predict how 250, 500 and 900 MHz trade resolution for penetration. Record λ/4 and the deepest visible event at each frequency. Write one sentence explaining why the highest frequency is not automatically the best choice.

METHOD 02 · MAGNETIC

Magnetic survey

A magnetic survey maps small perturbations of Earth's ambient field caused by susceptibility contrasts. This workspace superposes induced-dipole spheres and exact 3D prisms in a uniform inducing field.



Total-field anomaly ΔT(x) for the buried steel tank: a paired high–low pattern whose skew depends on field inclination. The dashed curve is the prism contribution; the solid curve adds deterministic noise.

SOLVER	SYNTHETIC OUTPUT	KEY UNITS
SEMI-ANALYTICAL	**ΔT profile + map**	**nT · m · SI**

WHAT YOU CONTROL

- **Ambient field F₀** (20 000–70 000 nT) and **inclination**; declination is fixed at 0°.
- **Body type**: sphere (dipole) or prism (finite strike); centre x, y, z, radius or x_1/x_2, z_1/z_2, strike length.
- **Susceptibility χ** (SI): the contrast that creates the induced moment.
- **Profile and map sampling**: line length, station spacing, number of map lines and line spacing.
- **Noise (nT) and seed**; body-contribution and station toggles.

WHAT YOU READ

- **ΔT profile**: a compact body gives a paired high–low anomaly; depth broadens it and reduces peak amplitude.
- **Anomaly map**: filled contours over the survey grid; the star marks the maximum |ΔT| and the dashed line is y = 0.
- **Diagnostics**: peak |ΔT|, RMS, peak-to-peak, station count and body moment.
- **Shape**: prism edges produce sharper gradients; a sphere is a far-field dipole. Skew changes with inclination.
- **Warnings**: strong χ, near-surface prisms, a sphere intersecting the surface, or a dipole whose radius exceeds its depth.

BEST PRACTICE

Change one thing: first hold depth and vary χ, then hold χ and vary depth. Compare the profile and map together — a map can reveal a strike direction that a single profile hides. Keep the body inside the line and read the warnings before interpreting amplitude.

LIMITS TO STATE

Induced magnetization only: no remanence, self-demagnetization, magnetic interaction, terrain or host variation. Spheres are point dipoles; prisms use exact corner formulas with finite strike. ΔT uses the small-anomaly approximation. The map is noise-free; no inversion is performed.

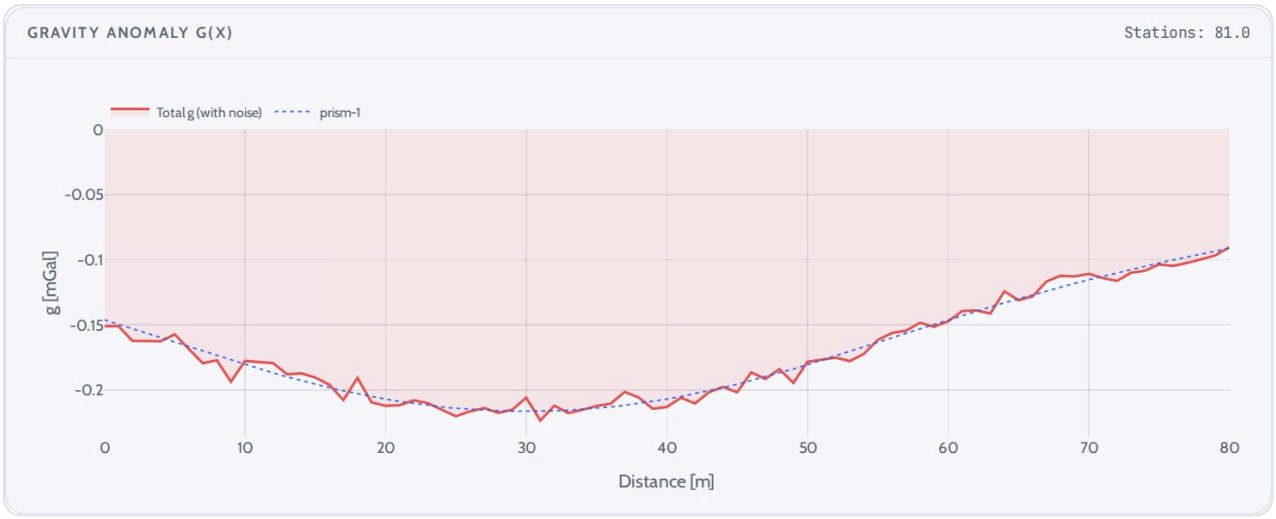
Exercise · Contrast, not just size

Run a small sphere with high χ, then a larger sphere with low χ. Match the peak amplitudes and compare the anomaly widths. Explain why peak amplitude alone cannot separate size from susceptibility, and which second observation you would need in the field.

METHOD 03 · GRAVITY

Gravity survey

Gravity measures the vertical attraction of density contrasts. The workspace uses exact 3D closed forms for spheres and rectangular prisms, so you can separate the effects of depth, size and density contrast.



Gravity anomaly g(x) for a low-density body. The broad negative curve is the model response; the noisier red line adds deterministic Gaussian noise. Read the shape and width before the peak value.

SOLVER	SYNTHETIC OUTPUT	KEY UNITS
ANALYTICAL	g profile + map	mGal · m · kg/m³

WHAT YOU CONTROL

- **Body type**: sphere (point mass) or prism (finite strike), with centre and radius or x_1/x_2, z_1/z_2 and strike length.
- **Density contrast Δρ** (kg/m³): relative to the host; negative for a cavity or low-density body.
- **Profile and map sampling**: line length, station spacing, map lines and line spacing.
- **Noise (mGal) and seed**; body-contribution and station toggles.
- **Multiple bodies**: add or remove prisms to test superposition.

WHAT YOU READ

- **g profile** (mGal): a smooth, broad anomaly. Negative for a mass deficit, positive for an excess.
- **Anomaly map**: filled contours with the maximum |g| star and the y = 0 profile line.
- **Summary**: peak-to-peak, peak g, excess mass and body parameters.
- **Width versus amplitude**: depth controls breadth more than peak; volume controls amplitude strongly.
- **Warnings**: near-surface prisms, thin shallow slabs, a sphere intersecting the surface, or radius exceeding depth.

BEST PRACTICE

Interpret the residual anomaly, not the raw field. Change depth first, then radius or Δρ, and compare both the profile and the map. Use the summary to check that your intuition and the model agree. Convert units explicitly: the solver works in SI and displays mGal.

LIMITS TO STATE

Constant density per body; flat ground; no terrain, tidal, free-air, Bouguer or regional/isostatic correction. Spheres are point masses; prisms use exact 3D corner formulas. Contributions superpose linearly. Non-uniqueness is fundamental; the map is noise-free.

Exercise · Radius versus amplitude

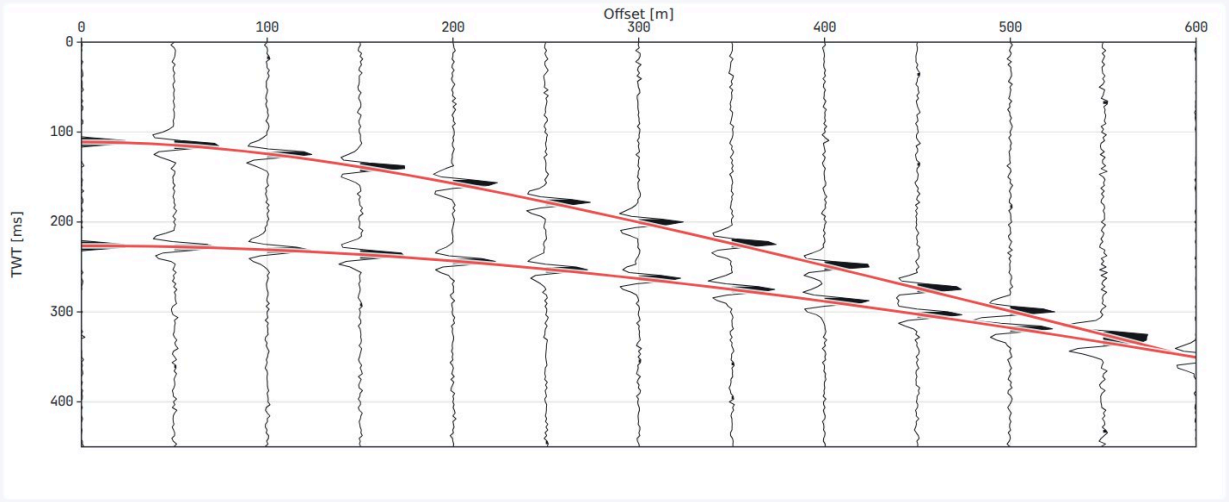
Run a sphere with radius 1 km and record the peak. Double the radius at the same depth and Δρ. Predict the ratio using the volume argument, then compare the measured ratio and the anomaly width. Explain why a deeper body can produce a broader but weaker anomaly.

METHOD 04 · REFLECTION

Seismic reflection

Reflection seismics images interfaces where acoustic impedance Z = ρ·Vp changes. The workspace computes a 1D primary-only reflectivity series, convolves it with a Ricker wavelet, and adds an NMO gather for offset analysis.

OFFSET GATHER (NMO)



Field-style variable-area display: black wiggle traces with positive lobes filled; red curves overlay the interpreted NMO moveout of each pick. Amplitudes are schematic (no geometric spreading or AVO).

Offset gather for the sand–shale sequence. Black wiggle traces are the synthetic traces; the red curves are the interpreted NMO moveout of each pick. Relative amplitudes are schematic — no geometric spreading or AVO.

SOLVER	SYNTHETIC OUTPUT	KEY UNITS
ANALYTICAL · PRIMARIES	Trace + NMO gather	ms · m · Hz

WHAT YOU CONTROL

- **Layer stack**: thickness, Vp and density ρ for each finite layer and the half-space.
- **Ricker frequency** (5–200 Hz): controls wavelet length and λ/4 resolution.
- **Gather offsets**: number of offsets, minimum and maximum offset; automatic or manual time window.
- **Noise (%) and seed**: deterministic trace noise.
- **Display**: NMO curves, wiggle/variable-area, grid and units.

WHAT YOU READ

- **Zero-offset trace**: two-way time, polarity and relative amplitude of each primary.
- **Reflection coefficient** $R = (Z_2-Z_1)/(Z_2+Z_1)$. A negative R reverses the wavelet polarity.
- **Gather**: hyperbolic NMO; red curves use $t^2 = t_0^2 + x^2/Vrms^2$. The curvature carries Vrms.
- **Picks table**: depth, TWT, impedance above/below, R, cumulative transmission and relative amplitude.
- **Diagnostics**: sample interval, dominant wavelength, λ/4 resolution and warnings for thin beds or wide angles.

BEST PRACTICE

Read polarity before brightness. Keep the wavelet and time window fixed when comparing layers, and use the picks table to connect the model to the trace. If two reflections merge, you are below the tuning thickness — do not report two separate interfaces. A bright event is an impedance contrast, not automatically gas.

LIMITS TO STATE

1D flat layers, primaries only; no multiples, mode conversions, absorption, anisotropy or AVO. The gather applies NMO without stretch, muting or geometric spreading. The source is a zero-phase Ricker wavelet. Amplitudes are normalized and schematic; no inversion is performed.

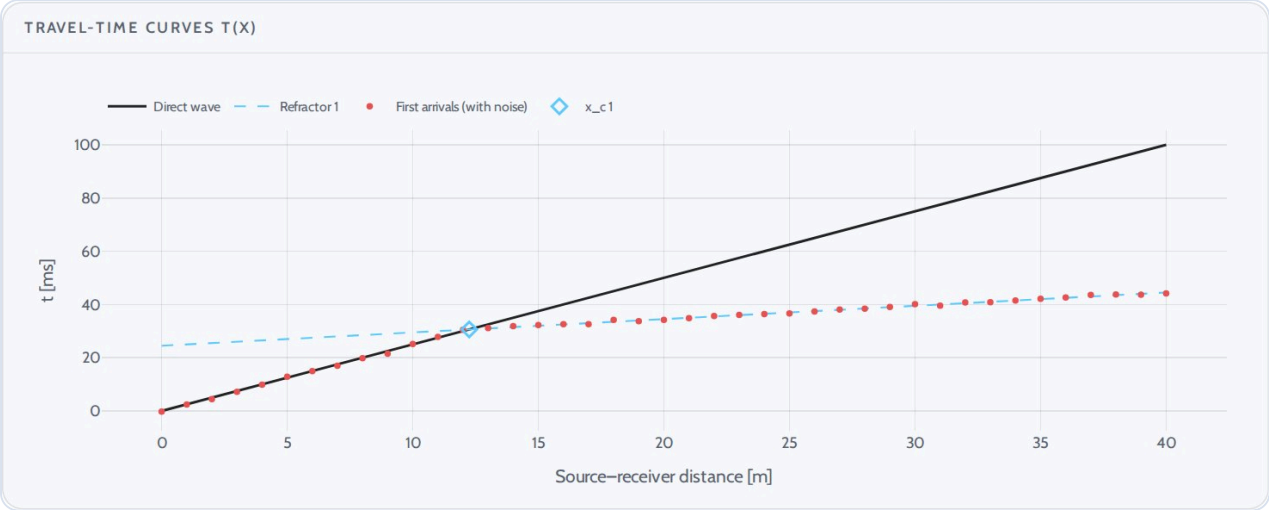
Exercise · Polarity, not just brightness

Load `#/simulator/reflection?scenario=gasSand` and locate the top and base of the gas sand. Compare its top wavelet with a positive-contrast event from the hard-bottom scenario. Relate the sign of R to the impedance change and explain why a bright spot alone is not proof of gas.

METHOD 05 · REFRACTION

Seismic refraction

Refraction uses the first seismic arrivals to estimate layer velocities and depths. The workspace computes direct and head-wave branches, their crossover distances, and a synthetic shot record for a layered model.



Travel-time curves t(x): the direct wave, one refractor branch, noisy first arrivals and the crossover marker. The slope of each branch is the inverse of its velocity; the intercept carries depth.

SOLVER	SYNTHETIC OUTPUT	KEY UNITS
ANALYTICAL · HEAD WAVES	t(x) + shot record	ms · m · m/s

WHAT YOU CONTROL

- **Layer stack**: thickness and Vp for each finite layer and the half-space.
- **Spread length and geophone spacing**: the geometry that decides which layers can become first arrivals.
- **Source frequency**, **pick noise** (ms) and **trace noise** (%): deterministic and seeded.
- **Display**: wiggle traces, AGC, ray paths and the velocity heatmap alternative.
- **Add/remove layers** to test increasing, decreasing and hidden-layer cases.

WHAT YOU READ

- **t(x) curve**: direct-wave slope $1/V_1$; head-wave slope $1/V_2$; the crossover is where the head wave becomes first.
- **First arrivals** versus clean branches: noise moves picks, not the underlying model.
- **Depth estimate**: the classic two-layer intercept-time formula for the first refractor only.
- **Shot record**: one Ricker-waved trace per geophone; red lines mark each branch where it is the first arrival.
- **Warnings**: velocity inversion/hidden layer, no refractor, or a spread too short to reach the crossover.

BEST PRACTICE

Make sure the spread passes the crossover before interpreting a refractor. For a dipping interface, run forward and reverse profiles — a single direction confuses dip with velocity. Pick first breaks consistently, use dense spacing near shallow targets, and never invent a low-velocity layer that never appears as a first arrival.

LIMITS TO STATE

Layered medium, constant velocity per layer, planar horizontal interfaces, flat topography. Only direct and head waves are computed; reflections, turning rays, attenuation, anisotropy and lateral variation are absent. Velocity inversions create hidden layers. The depth estimate is a two-layer aid, not a general inversion.

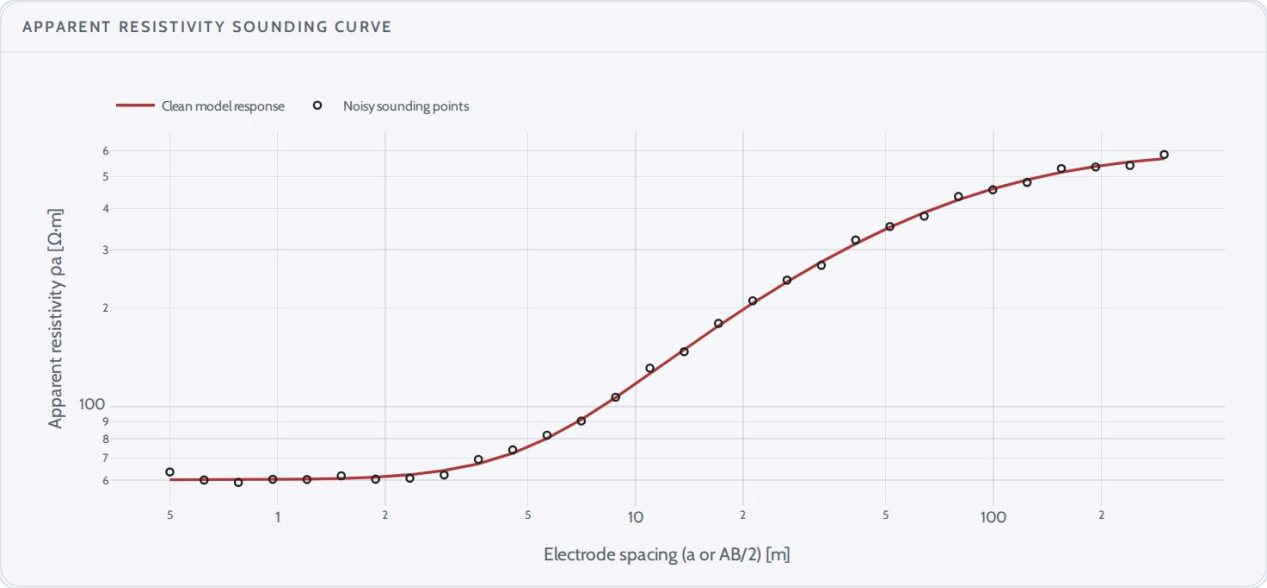
Exercise · The hidden layer

Compare `#/simulator/refraction?scenario=threeLayer` with `?scenario=velocityInversion`. Thin the middle layer until its branch disappears from the first arrivals. Explain why the deeper refractor can still be detected while the slow layer cannot.

METHOD 06 · RESISTIVITY

Resistivity sounding (VES)

Vertical electrical sounding injects direct current and measures apparent resistivity as electrode spacing grows. The workspace solves the 1D layered-earth Hankel integral for a Wenner or Schlumberger array.



Apparent-resistivity sounding curve for the two-layer bedrock model: a low-resistivity soil plateau, then a rising branch toward the resistive half-space. The circles add deterministic multiplicative noise.

SOLVER	SYNTHETIC OUTPUT	KEY UNITS
NUMERICAL · 1D	ρa curve + model	Ω·m · m

WHAT YOU CONTROL

- **Array**: Wenner (equal spacing) or Schlumberger (fixed MN, expanding AB).
- **Spacing**: minimum, maximum and number of soundings; MN/2 for Schlumberger geometry.
- **Layer stack**: 2–8 layers, thickness and resistivity; the last layer is a half-space.
- **Noise (%) and seed**: multiplicative log-normal noise for display.
- **Display/export**: clean model response and noisy sounding points; CSV and PNG.

WHAT YOU READ

- **ρa curve** on log–log axes: flat means uniform; ascending/descending branches mark resistivity contrasts.
- **Layered model and electrode array**: connect the geometry to the depth scale.
- **Summary**: top and bottom resistivity, total depth and ρa range.
- **Equivalence**: a thin strong contrast and a thick weak contrast can fit the same curve.
- **Warnings/errors**: Schlumberger geometry, layer count, total depth and spacing order.

BEST PRACTICE

Use logarithmic spacing and extend the maximum spacing beyond the target depth. Compare Wenner and Schlumberger on the same model before comparing field arrays. Read the curve's shoulders and branch positions, not just the last point. State equivalence explicitly when two layer models fit.

LIMITS TO STATE

1D horizontal layers, infinite lateral extent, direct current. No induced polarization, anisotropy, topography, lateral change or electrode contact resistance. The Hankel integral is truncated and quadrature-based; extreme contrasts may need finer sampling. Noise is display-only; no inversion is performed.

Exercise · Wenner versus Schlumberger

Run `#/simulator/resistivity?scenario=two-layer-bedrock` with Wenner, then switch to Schlumberger without changing the layers. Compare curve shape and depth sensitivity. Explain why “the same ground” can produce two different apparent-resistivity curves.

METHOD 07 · HVSR FORWARD

HVSR forward simulation

HVSR uses ambient vibrations to estimate site resonance. This workspace computes a 1D layered SH transfer function and a Nakamura-style H/V proxy, then reports f0, A0 and the quarter-wave depth estimate.



H/V spectral ratio for the IKN alluvium preset. The dashed marker is f0 near the quarter-wave estimate; a larger higher-mode peak exists at about 4.2 Hz. The warning is part of the result, not an error to hide.

SOLVER	SYNTHETIC OUTPUT	KEY UNITS
PASSIVE · 1D PROXY	H/V curve + Vs profile	Hz · m/s · m

WHAT YOU CONTROL

- **Layer stack**: thickness and Vs for each finite layer and the half-space. Density and Vp/Vs are fixed.
- **Frequency band**: minimum, maximum and number of frequency samples.
- **Shear Q**: numerical damping that broadens and lowers the resonance.
- **Noise (%) and seed**: deterministic multiplicative noise.
- **Presets and export**: four teaching columns; H/V CSV and model CSV.

WHAT YOU READ

- **H/V curve**: f0 is the fundamental marker near the quarter-wave estimate; A0 is its amplitude.
- **Quarter-wave estimate** and total depth: a first-order depth check, not a unique profile.
- **Vs profile**: the layered model that generated the curve.
- **Warnings**: higher-mode dominance, peak at the band edge, or weak impedance contrast.
- **Compare/sweep**: A/B curves and one-factor sensitivity of f0 and A0.

BEST PRACTICE

Use a band wide enough to see the peak and at least one higher mode. Do not choose the tallest peak blindly: check whether f0 is consistent with the quarter-wave depth. Keep density and Vp/Vs fixed in your reasoning, and treat the curve as a proxy — not a full Rayleigh ellipticity calculation.

LIMITS TO STATE

1D layered Earth, constant Vs and density per layer. Vertically propagating SH waves; H/V approximated by horizontal transfer with unit vertical input. Fixed Vp/Vs = 1.8 and density = 2000 kg/m³. No Rayleigh ellipticity, higher modes, lateral heterogeneity, topography or anisotropy. Synthetic only; no field microtremor data.

Exercise · Thickness sets the resonance

Load `#/simulator/hvsr?scenario=iknAlluvium` and record f0. Double the thickness of the soft layers, predict the new f0 from f0 ≈ Vs/(4H), then run and compare. Explain why the curve alone cannot tell you whether the layer became thicker or slower.

HVSR INVERSION

What the curve constrains — and what it does not

The HVSR workspace adds one clearly labelled inversion experiment. It searches for layered Vs profiles whose H/V curves match the synthetic target, then shows the ensemble instead of pretending there is one answer.



Inversion result for the IKN alluvium target: the best-fit curve (red) tracks the target (blue), while the Vs ensemble (violet band) shows where the profile is and is not constrained. The low misfit is not proof of a unique model.

1 · Set the search space

Choose the inversion layer count, initial samples, best models kept, new samples per iteration, iterations and seed.

2 · Run and read the fit

The best-fit curve is compared with the synthetic target. Misfit and evaluation count are diagnostics, not a geological score.

3 · Read the ensemble

The Vs ensemble (5–50–95%) is the honest result. A narrow band at a depth means the curve constrains it; a wide band means it does not.

What the search is doing

A neighborhood-algorithm-style search samples a bounded layered space, keeps the best-fitting models, and refines around them. It is inspired by published HVSR practice but is not a Geopsy reimplementation.

More inversion layers can fit the curve more flexibly — and make the profile less unique. That trade-off is the lesson.

How to read the result

- Target H/V versus best-fit H/V: check peak positions, amplitudes and the shape between peaks.
- Misfit: a normalized curve error. Lower is better only within the same target and settings.
- Model evaluations: more search does not guarantee a unique answer.
- Best-fit Vs profile versus the 5–95% band: report the band, not the single line.
- Stale warning: if the forward model or settings changed, rerun before comparing.

THE HONESTY RULE

A low misfit does not prove a unique or geologically correct profile. The 5–95% envelope is the result; the best line is one ensemble member. The target itself is synthetic and generated from the model you see — no field data are inverted.

Exercise · Four layers, then six

Run the inversion with 4 layers, note the misfit and the width of the Vs band. Rerun with 6 layers. Did the fit improve? Did the band narrow or widen? Write two sentences separating “better fit” from “better constrained”.

EXPERIMENT TOOLS

Compare, sweep and challenge

A single run is a demonstration. These three tools turn it into a controlled experiment — provided you keep one variable and one honest question at a time.

A/B compare

Pin the current result as A. Change one control, run again, and read the deltas in the metrics and the overlaid curves. A stays fixed until you replace or clear it.

One-factor sweep

Choose one parameter, a range and a number of steps. The solver recomputes each step with noise muted, so the trend reflects the model, not the seed.

Challenge

A hidden target is generated from a nearby model. Tune your model, check the score, and compare the curve shape. The score measures similarity, not geological truth.



Challenge panel after a GPR attempt: the hidden target (blue) and the current model (red) differ in arrival time, so the score is 78/100 and the curve mismatch is 11.2%. The feedback names what to compare next.

QUESTION	TOOL	OUTPUT TO RECORD	DO NOT CONCLUDE
Did changing X alter Y?	A/B compare	Metric deltas, overlaid curves	That X is the only possible cause
How sensitive is Y to X?	Sensitivity sweep	Trend, breakpoint, guarded range	A formal uncertainty estimate
Can I recover the hidden model?	Challenge	Score, mismatch, feature differences	That one fit is the true model
Can I defend the interpretation?	Report + evidence pack	Permalink, figure, CSV, limitation	That a screenshot is enough

DO NOT GAME THE SCORE

The challenge is a curve-matching exercise. Several models can fit the same curve; a high score is evidence of a consistent response, not a unique geological solution. Record the parameters you changed and one alternative model that would also fit.

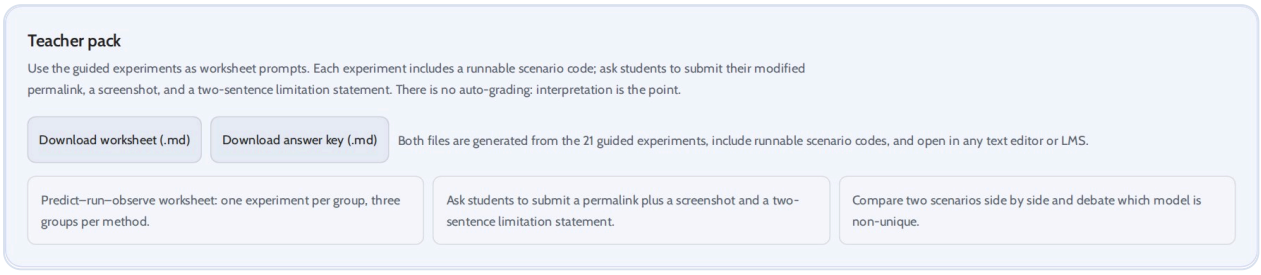
Exercise · Three tools, one question

Use GPR. Pin 400 MHz as A, run 900 MHz as B, then sweep 200–1200 MHz. Finally start a challenge and try to recover the hidden target. Write which tool gave you the clearest evidence and which one was easiest to misread.

CLASSROOM & SELF-STUDY

Teach the loop, not the button

A reproducible lesson plan, group roles, a simple rubric, and a self-study path that uses all seven methods — not only HVSR.



The teacher pack generates a worksheet and answer key from the 21 guided experiments. Each experiment carries a runnable scenario code and an expected observation.

A 55-minute lesson

0–5 min
Hook
Play the live HVSR demo. Ask what controls the peak.

5–15 min
Concept
Read one theory chapter and the solver note together.

15–30 min
Experiment
One preset per group; predict, run, change one control.

30–40 min
Compare
A/B or sweep, then a challenge round.

40–50 min
Evidence
Print the report and write the limitation sentence.

50–55 min
Exit ticket
One minute: what changed, why, and what remains uncertain.

Group roles

- **Driver** — operates the controls and shares the screen.
- **Recorder** — writes prediction, parameters and observation.
- **Skeptic** — names one alternative model and one limit.
- **Reporter** — presents the evidence pack in 60 seconds.

Assessment rubric

CRITERION	WHAT STRONG WORK LOOKS LIKE
Physical reasoning	Names the measured property and predicts the direction of change with a reason.
Controlled comparison	Changes one variable; keeps scenario, geometry and noise fixed.
Evidence	Submits a permalink, figure/CSV and the exact model values.
Limits	States what the model omits and one alternative that could fit.

Assignment that resists copying

Submit a modified permalink, one screenshot or CSV, and a two-sentence limitation statement. The permalink restores the exact model, so the evidence is reproducible.

Ask each group to choose a different experiment from the 21 — three per method. No auto-grading: interpretation is the point.

Self-study, one term

1. One method per week; read theory chapters 1 and 5.
2. Run all three guided experiments for that method.
3. Do one A/B comparison, one sweep and one challenge.
4. Build a one-page evidence pack and revise after feedback.

TEACHER PACK

Download the worksheet and answer key from `#/resources`. Both are generated from the 21 experiments, include runnable scenario codes, and open in any text editor or LMS.

EXPORT & EVIDENCE

Build an evidence pack, not a gallery

The app can export values, images, a report and a reproducible link. Use them together so another person can rerun your model and challenge your interpretation.

ACTION	WHAT YOU GET	WHEN TO USE IT
Share scenario	A URL that restores the exact method, scenario and model state (an opaque `?d=` token).	Reproducibility; paste into a report or LMS.
Export CSV	Plotted values for the current run; several workspaces include scenario and parameter metadata.	Analysis in a spreadsheet; checking numbers behind the chart.
Export PNG	The chart as a high-resolution image, without the app interface.	Slides, lab reports, posters.
Print / save PDF	The practice report: model, figures, interpretation prompts, name and date fields.	Submission and class discussion.
H/V CSV + model CSV	Frequency, H/V and SH values; layer thickness and Vs for the forward model.	Checking the curve and the model outside the app.
Teacher pack	Worksheet and answer key in Markdown, generated from the 21 experiments.	Lesson planning and assessment.

The evidence pack

- ✓ Method, scenario and exact parameters
- ✓ One figure or CSV of the response
- ✓ One alternative model
- ✓ Date and app version
- ✓ Permalink that restores the run
- ✓ Prediction written before the run
- ✓ One limitation sentence
- ✓ Name / group for submission

File names that survive

```
kb-gpr-sandBedrock-400MHz-2026-11-01.png
kb-hvsr-iknAlluvium-fwd-curve.csv
kb-hvsr-iknAlluvium-inv-ensemble.png
kb-report-gpr-group3-2026-11-01.pdf
```

Use method, scenario, key parameter and date. Avoid “final”, “final2” and screenshots named IMG_4821.

What a permalink contains

The route names the method: `#/simulator/gpr`. The `?d=` token encodes the scenario and the full model. A readable `?scenario=sandBedrock` link opens only the preset, which is what the teacher pack uses.

Do not edit the token by hand. If a link is too long for an LMS, submit the scenario code plus a screenshot and the parameter table instead.

Report prompts

1. What changed in the synthetic response when you modified the model?
2. Which parameter most strongly controls the anomaly or curve, and why?
3. Which model assumption most limits your interpretation?

Answer in full sentences. A number without a reason is not evidence.

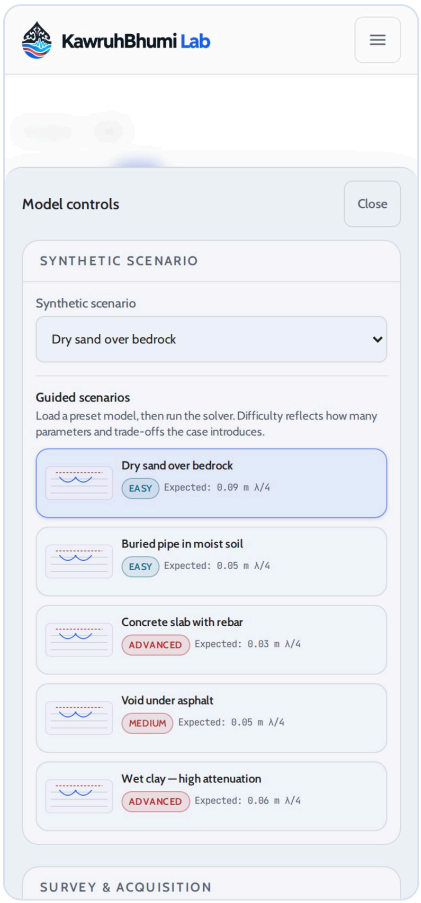
SUBMIT THE MODEL, NOT JUST THE PICTURE

A screenshot can be beautiful and still be unreproducible. The permalink and the parameter values are what let a teacher or reviewer check your reasoning.

MOBILE & LOW BANDWIDTH

A smaller screen, the same physics

At 390 px the layout changes, not the solver. The controls move into a sheet, the run bar stays sticky, and the chart keeps its fullscreen toolbar.



390 px: the Controls sheet keeps presets, scenario and model fields readable. The run bar stays sticky above it.

Mobile workflow

1. Open the method route, then tap **Controls**. The model sheet slides up without hiding the run bar.
2. Load a preset, edit one value, and close the sheet. The sticky bar keeps **Run Simulation** within reach.
3. Scroll to the chart; use the chart toolbar for fullscreen, zoom and reset. Read the diagnostics below the chart.
4. Tap **Share scenario** to copy a reproducible link before you leave the page.
5. Export CSV or PNG while the result is current. A stale result is not evidence.

Low bandwidth

- The app is client-side: after the first load, solver calculations run locally. Keep the tab open rather than reloading.
- Reduce trace count, frequency samples or time window before blaming the network. Charts are the heavy part.
- Turn off grid and station dots for faster rendering on a low-end phone.
- Download the PDF guide once and keep it offline; it is small and self-contained.

Touch targets

Buttons are at least 44 px tall. Use the chart fullscreen control instead of pinch-zooming the whole page.

One method at a time

A phone is not a workstation. Load one workspace, run it, export, then switch method from the Method switcher.

No account needed

The model is stored in the URL and local storage. There is no upload of your scenario; sharing is explicit.

MOBILE CHECKLIST

Preset loaded · one control changed · result current · warning read · CSV/PNG saved · permalink copied. If one item is missing, the phone has not finished the experiment.

TROUBLESHOOTING

When the chart will not cooperate

Most problems are model, input or browser issues — not a broken solver. Work through the symptom table, then the preflight checklist.

SYMPTOM	LIKELY CAUSE	WHAT TO DO
Chart is empty	No current result, or invalid input.	Check the error list, fix the field, press Run.
"Results are outdated"	The model changed after the last run.	Press Run again before reading or exporting.
High-attenuation warning	The physics, not a bug: conductive ground absorbs the signal.	Change the model or report the limitation; do not hide it.
Hidden layer / no refractor	A velocity inversion or a spread shorter than the crossover.	Lengthen the spread or state that the layer is invisible.
Inversion button disabled	Forward result missing, stale, or invalid.	Run and validate the forward model first.
Scenario link opens default	Wrong method route or a mistyped `?scenario=` code.	Use the full code from the teacher pack; check the method.
Share link is very long	The full model is encoded in the token.	Expected. Submit the scenario code and parameter table if needed.
Download blocked	Browser download permission or a private window.	Allow downloads, retry, or copy the values from the chart.
Print is blank or partial	Charts render asynchronously; print was pressed too early.	Run, wait for the chart, then print; or use PNG/CSV.
Phone feels slow	Too many traces, samples or map cells.	Reduce resolution, turn off grid, close other tabs.

Ten-minute preflight

- ✓ Method route matches the question
- ✓ Inputs valid; no red errors
- ✓ Axes, units and scale understood
- ✓ One control changed at a time
- ✓ CSV/PNG and permalink saved
- ✓ Preset and scenario name recorded
- ✓ Result current, not stale
- ✓ Warnings read and explained
- ✓ Prediction written before the run
- ✓ Limitation sentence drafted

When to stop and ask

Stop when a warning says the model is outside its supported range, when two methods give incompatible answers, or when you cannot name the measured property. Do not force a curve to look right by adding noise or changing several controls.

When asking for help, include: method, scenario, permalink, screenshot, the exact parameter values, and the warning text.

kholis.nh3@gmail.com
Corrections on physics, references and wording are welcome.

A WARNING IS PART OF THE RESULT

Do not hide a warning to make a figure look cleaner. The warning tells the reader which part of the model is least trustworthy — which is exactly what an honest interpretation must say.

REFERENCE

Glossary, symbols and the honest close

Use this page while reading a chart: name the symbol, its unit, and the assumption behind it. Precision in vocabulary prevents precision in the wrong place.

Glossary

GPR	Ground-penetrating radar; EM reflection profiling.	**A-scan**	Amplitude versus two-way time at one trace.
εr	Relative dielectric permittivity; controls EM velocity.	**σ**	Electrical conductivity; drives attenuation.
χ	Magnetic susceptibility (SI).	**ΔT**	Total-field magnetic anomaly (nT).
Δρ	Density contrast (kg/m³).	**mGal**	Gravity unit: 10^{-5} m/s².
Z	Acoustic impedance ρ·Vp.	**R**	Normal-incidence reflection coefficient.
NMO	Normal moveout: offset-dependent reflection time.	**Head wave**	Critically refracted wave along a faster layer.
Crossover	Offset where a head wave becomes the first arrival.	**VES**	Vertical electrical sounding.
ρa	Apparent resistivity (Ω·m).	**Equivalence**	Different layer models fitting the same curve.
HVSR	Horizontal-to-vertical spectral ratio.	**f0**	Fundamental resonance frequency (Hz).
A0	Amplification at f0.	**Vs**	Shear-wave velocity (m/s).
Misfit	Normalized curve error in the inversion.	**Ensemble**	Set of models consistent with the data.
Forward model	Model → synthetic response.	**Inversion**	Data → candidate model(s); non-unique.

Symbols and formulas

$v = c/\sqrt{\varepsilon r}$	EM velocity in a low-loss dielectric.
$z = v \cdot t/2$	Depth from two-way time.
$R = (Z_2 - Z_1)/(Z_2 + Z_1)$	Normal-incidence reflection coefficient.
$t^2 = t_0^2 + x^2/Vrms^2$	NMO moveout.
$f0 \approx Vs/(4H)$	Quarter-wave resonance estimate.
$\rho a = 2\pi a \cdot \Delta V/I$	Wenner apparent resistivity (ideal array).

UNITS MATTER

The solver works in SI and displays mGal, nT, Ω·m, ms and ns. A unit conversion is a modelling decision: write it down.

One-sentence test

Can you say what was measured, what was modelled, what changed, and what remains uncertain — without looking at the screen? If not, the experiment is not finished.

Guide scope

This guide covers KawruhBhumi Lab v2.7.0: seven forward workspaces, 35 theory chapters, 21 guided experiments and one labelled HVSR inversion. It is a teaching companion, not a calibration manual.

Corrections

Found an error in the physics, a broken reference or an unclear assumption? Write to **kholis.nh3@gmail.com** with the method, page and scenario. Corrections make the lab more honest.

KawruhBhumi Lab · v2.7.0
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